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Multi-Qubit States Introduction

Video: Multi-Qubit States Introduction
5 min

Graded Assignment: Assessment 8
Submitted

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Next item

Assessment 8

Review Learning Objectives

1. Select the true statements about the transformations of the S gate. **1 / 1 point**

- [ ] The S gate is an identity matrix.
- [x] The S gate is also known as the phase gate.

Correct
Correct! The S gate is sometimes referred to as the phase gate.

- [x] The S gate can be represented as a rotation around the Z-axis by $\frac{\pi}{2}$.

Correct
Correct! The S gate can indeed be represented as a rotation around the Z-axis by $\frac{\pi}{2}$.

- [x] The S gate has a matrix representation of $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$.

Correct
Correct! The matrix representation of the S gate is $\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$.

- [ ] The S gate is equivalent to applying a T gate twice.

2. Which of the following properties is unique to the T gate? **1 / 1 point**

- ( ) It is equivalent to the H gate followed by a Z gate.
- (•) It produces a $\frac{\pi}{4}$ phase shift.
- ( ) It transforms into the identity matrix when squared.
- ( ) It produces a $\frac{\pi}{2}$ phase shift.

Correct
Well done! The T gate is unique in that it produces a $\frac{\pi}{4}$ phase shift.

3. Which quantum gate is known as the identity gate? **1 / 1 point**

- ( ) U1 gate
- ( ) U3 gate
- (•) I gate
- ( ) U2 gate

Correct
Correct! The I gate is known as the identity gate because it leaves the qubit unchanged.

4. How does entanglement benefit quantum computations? **1 / 1 point**

- ( ) It simplifies the implementation of quantum gates.
- (•) It allows for the correlation of qubit states that can be used for complex computations.
- ( ) It increases the error rates, making quantum computations more reliable.
- ( ) It speeds up the processing time of quantum algorithms.

Correct
Correct! Entanglement enables complex correlations which are useful for quantum computations.

5. Which of the following best explains why multiple qubits are needed in quantum computing? **1 / 1 point**

- ( ) To increase the speed of quantum computations exponentially.
- ( ) To reduce the error rates in quantum computations.
- ( ) To simplify the control and manipulation of quantum gates.
- (•) To represent and process complex quantum states that cannot be handled by a single qubit.

Correct
Correct! Multiple qubits allow for the representation and processing of complex quantum states, enabling more advanced computations.

coach
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Assignment details
Due Dec 20, 11:59 PM IST
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